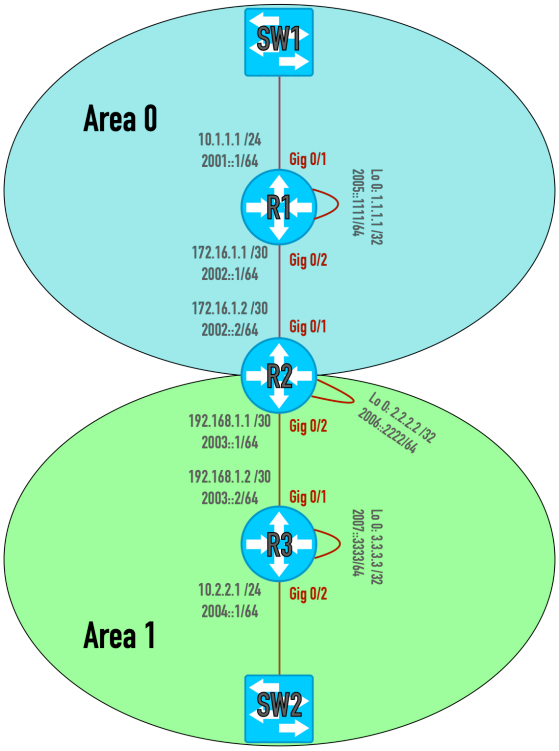
OSPFv3 Traditional Configuration

Topology



Initial Configuration Commands

R1:

```
enable
conf t
host R1
no banner motd
no banner login
no banner exec
no banner incoming
line vty 0 15
password cisco
login
exec-timeout 0 0
line con 0
logging synchronous
no ip-domain lookup
int gig 0/1
ip address 10.1.1.1 255.255.255.0
```

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```
ipv6 address 2001::1/64
no shut
int gig 0/2
ip address 172.16.1.1 255.255.255.252
ipv6 address 2002::1/64
no shut
int lo0
ip address 1.1.1.1 255.255.255.255
ipv6 address 2005::1111/64
end
copy run star
```

R2:

```
enable
conf t
host R2
no banner motd
no banner login
no banner exec
no banner incoming
line vty 0 15
password cisco
login
exec-timeout 0 0
line con 0
logging synchronous
no ip-domain lookup
int lo0
ip address 2.2.2.2 255.255.255.255
ipv6 address 2006::2222/64
int gig 0/1
ip address 172.16.1.2 255.255.255.252
ipv6 address 2002::2/64
no shut
int gig 0/2
ip address 192.168.1.1 255.255.255.252
ipv6 address 2003::1/64
no shut
end
copy run star
```

R3:

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```
enable
conf t
host R3
no banner motd
no banner login
no banner exec
no banner incoming
line vty 0 15
password cisco
login
exec-timeout 0 0
line con 0
logging synchronous
no ip-domain lookup
int lo0
ip address 3.3.3.3 255.255.255.255
ipv6 address 2007::3333/64
int gig 0/1
ip address 192.168.1.2 255.255.255.252
ipv6 address 2003::2/64
no shut
int gig 0/2
ip address 10.2.2.1 255.255.255.0
ipv6 address 2004::1/64
no shut
end
copy run star
```

Lab Tasks

1. Let's start by configuring OSPFv2 on all of our routers to support our IPv4 networks.
2. On R3, let's make sure that R3 knows about all of the networks within our topology. Let's see if we are also able to ping R1's loopback interface.
3. Let's go into R1 and enable IPv6, as well as set up OSPFv3 for each of our interfaces.
4. Let's go into R2 and enable IPv6, as well as set up OSPFv3 for each of our interfaces.
5. Let's go into R3 and enable IPv6, as well as set up OSPFv3 for each of our interfaces.
6. On R3, let's verify that we can see the IPv6 networks by using the "show ipv6 route" command. Verify that we can see our IPv6 neighbor. Lastly, let's look at interface gig 0/1.
7. On R3, take a look at the IPv6 OSPF database.

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Solution

Step 1: Let's start by configuring OSPFv2 on all of our routers to support our IPv4 networks.

```
R1>en
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#router ospf 1
R1(config-router)#network 0.0.0.0 255.255.255.255 area 0
R1(config-router)#end

R2>en
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#router ospf 1
R2(config-router)#network 172.16.1.0 0.0.0.3 area 0
R2(config-router)#network 192.168.1.0 0.0.0.3 area 1
R2(config-router)#network 2.2.2.2 0.0.0.0 area 1
R2(config-router)#end

R3>en
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#router ospf 1
R3(config-router)#network 192.168.1.0 0.0.0.3 area 1
R3(config-router)#network 3.3.3.3 0.0.0.0 area 1
R3(config-router)#int gig 0/2
R3(config-if)#ip ospf 1 area 1
R3(config-if)#end
```

Step 2: On R3, let's make sure that R3 knows about all of the networks within our topology. Let's see if we are also able to ping R1's loopback interface.

```
R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static
route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set
```

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```
      1.0.0.0/32 is subnetted, 1 subnets
O IA     1.1.1.1 [110/3] via 192.168.1.1, 00:04:25, GigabitEthernet0/1
      2.0.0.0/32 is subnetted, 1 subnets
O        2.2.2.2 [110/2] via 192.168.1.1, 00:04:25, GigabitEthernet0/1
      3.0.0.0/32 is subnetted, 1 subnets
C        3.3.3.3 is directly connected, Loopback0
      10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
O IA     10.1.1.0/24 [110/3] via 192.168.1.1, 00:04:25, GigabitEthernet0/1
C        10.2.2.0/24 is directly connected, GigabitEthernet0/2
L        10.2.2.1/32 is directly connected, GigabitEthernet0/2
      172.16.0.0/30 is subnetted, 1 subnets
O IA     172.16.1.0 [110/2] via 192.168.1.1, 00:04:25, GigabitEthernet0/1
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.1.0/30 is directly connected, GigabitEthernet0/1
L        192.168.1.2/32 is directly connected, GigabitEthernet0/1
R3#ping 1.1.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 1.1.1.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 3/5/9 ms
```

(##Notice how we have knowledge of each of the loopback addresses of each of our routers within R3's routing table.)

Step 3: Let's go into R1 and enable IPv6, as well as set up OSPFv3 for each of our interfaces.

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ipv6 unicast-routing
R1(config)#ipv6 cef
R1(config)#ipv6 router ospf 1
R1(config-rtr)#router-id 1.1.1.1
R1(config-rtr)#passive-interface lo0
R1(config-rtr)#passive-interface gig 0/1
R1(config-rtr)#?
  adjacency              Control adjacency formation
  area                   OSPF area parameters
  auto-cost              Calculate OSPF interface cost according to bandwidth
  bfd                    BFD configuration commands
  compatible             Compatibility list
  default                Set a command to its defaults
  default-information    Distribution of default information
  default-metric         Set metric of redistributed routes
  discard-route          Enable or disable discard-route installation
  distance               Administrative distance
  distribute-list        Filter networks in routing updates
  event-log              Event Logging
  exit                   Exit from IPv6 routing protocol configuration mode
  graceful-restart       Graceful-restart options
  interface-id           Source of the interface ID
  limit                  Limit a specific OSPF feature
  local-rib-criteria     Enable or disable usage of local RIB as route criteria
  log-adjacency-changes  Log changes in adjacency state
  manet                  Specify MANET OSPF parameters
```

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```
  max-lsa                 Maximum number of non self-generated LSAs to accept
  max-metric              Set maximum metric
  maximum-paths           Forward packets over multiple paths
  no                      Negate a command or set its defaults
  passive-interface       Suppress routing updates on an interface
  prefix-suppression      Enable prefix suppression
  queue-depth             Hello/Router process queue depth
  redistribute            Redistribute IPv6 prefixes from another routing
                          protocol
  router-id               router-id for this OSPF process
  shutdown                Shutdown the router process
  snmp                    Modify snmp parameters
  summary-prefix          Configure IPv6 summary prefix
  table-map               Map external entry attributes into routing table
  timers                  Adjust routing timers

R1(config-rtr)#int gig 0/1
R1(config-if)#ipv6 ospf 1 area 0
R1(config-if)#int gig 0/2
R1(config-if)#ipv6 ospf 1 area 0
R1(config-if)#int lo0
R1(config-if)#ipv6 ospf 1 area 0
R1(config-if)#end
```

(##Unlike IPv4, IPv6 has to be enabled on a router to be able to use it.)
(##It is a good practice to turn on Cisco Express Forwarding because it makes the forwarding of packets a lot more efficient.)
(##Notice how there isn't a network command when it comes to IPv6. We must go into the interface to tell it that we want it to participate in OSPFv3.)

Step 4: Let's go into R2 and enable IPv6, as well as set up OSPFv3 for each of our interfaces.

```
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#ipv6 unicast-routing
R2(config)#ipv6 cef
R2(config)#ipv6 router ospf 1
R2(config-rtr)#router-id 2.2.2.2
R2(config-rtr)#int gig 0/1
R2(config-if)#ipv6 ospf 1 area 0
R2(config-if)#int gig 0/2
R2(config-if)#ipv6 ospf 1 area 1
R2(config-if)#int lo0
R2(config-if)#ipv6 ospf 1 area 1
R2(config-if)#end
```

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Step 5: Let's go into R3 and enable IPv6, as well as set up OSPFv3 for each of our interfaces.

```
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#ipv6 unicast-routing
R3(config)#ipv6 cef
R3(config)#ipv6 router ospf 1
R3(config-rtr)#passive-interface gig 0/2
R3(config-rtr)#passive-interface lo0
R3(config-rtr)#router-id 3.3.3.3
R3(config-rtr)#int gig 0/1
R3(config-if)#ipv6 ospf 1 area 1
R3(config-if)#int gig 0/2
R3(config-if)#ipv6 ospf 1 area 1
R3(config-if)#int lo0
R3(config-if)#ipv6 ospf 1 area 1
R3(config-if)#end
```

Step 6: On R3, let's verify that we can see the IPv6 networks by using the "show ipv6 route" command. Verify that we can see our IPv6 neighbor. Lastly, let's look at interface gig 0/1.

```
R3#show ipv6 route
IPv6 Routing Table - default - 11 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
       B - BGP, HA - Home Agent, MR - Mobile Router, R - RIP
       H - NHRP, I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea
       IS - ISIS summary, D - EIGRP, EX - EIGRP external, NM - NEMO
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       RL - RPL, O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1
       OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       la - LISP alt, lr - LISP site-registrations, ld - LISP dyn-eid
       lA - LISP away, a - Application
OI  2001::/64 [110/3]
     via FE80::5054:FF:FE12:99EB, GigabitEthernet0/1
OI  2002::/64 [110/2]
     via FE80::5054:FF:FE12:99EB, GigabitEthernet0/1
C   2003::/64 [0/0]
     via GigabitEthernet0/1, directly connected
L   2003::2/128 [0/0]
     via GigabitEthernet0/1, receive
C   2004::/64 [0/0]
     via GigabitEthernet0/2, directly connected
L   2004::1/128 [0/0]
     via GigabitEthernet0/2, receive
OI  2005::1111/128 [110/2]
     via FE80::5054:FF:FE12:99EB, GigabitEthernet0/1
O   2006::2222/128 [110/1]
     via FE80::5054:FF:FE12:99EB, GigabitEthernet0/1
C   2007::/64 [0/0]
     via Loopback0, directly connected
```

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```
L   2007::3333/128 [0/0]
     via Loopback0, receive
L   FF00::/8 [0/0]
     via Null0, receive
```

```
R3#show ipv6 ospf neighbor

            OSPFv3 Router with ID (3.3.3.3) (Process ID 1)

Neighbor ID     Pri   State           Dead Time   Interface ID    Interface
2.2.2.2           1   FULL/BDR        00:00:39    4
GigabitEthernet0/1
```

```
R3#show ipv6 ospf int gig 0/1
GigabitEthernet0/1 is up, line protocol is up
  Link Local Address FE80::5054:FF:FE0F:C1A2, Interface ID 3
  Area 1, Process ID 1, Instance ID 0, Router ID 3.3.3.3
  Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State BDR, Priority 1
  Designated Router (ID) 2.2.2.2, local address FE80::5054:FF:FE12:99EB
  Backup Designated router (ID) 3.3.3.3, local address
FE80::5054:FF:FE0F:C1A2
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:06
  Graceful restart helper support enabled
  Index 1/1/1, flood queue length 0
  Next 0x0(0)/0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 2
  Last flood scan time is 2 msec, maximum is 2 msec
  Neighbor Count is 1, Adjacent neighbor count is 1
    Adjacent with neighbor 2.2.2.2  (Designated Router)
  Suppress hello for 0 neighbor(s)
```

(##Notice how we can see the IPv6 loopback addresses of each of our routers.)
(##We can see that we are neighbors with R2.)
(##We can see that R3 is the backup designated router, and R2 is the designated router.)

Step 7: On R3, take a look at the IPv6 OSPF database.

```
R3#show ipv6 ospf database

            OSPFv3 Router with ID (3.3.3.3) (Process ID 1)

                Router Link States (Area 1)

ADV Router       Age         Seq#        Fragment ID  Link count  Bits
 2.2.2.2         222         0x80000004  0            1           B
```

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```
 3.3.3.3          174          0x80000005  0             1            None

                 Net Link States (Area 1)

ADV Router       Age          Seq#        Link ID    Rtr count
 2.2.2.2         221          0x80000001  4          2

                 Inter Area Prefix Link States (Area 1)

ADV Router       Age          Seq#        Prefix
 2.2.2.2         326          0x80000001  2005::1111/128
 2.2.2.2         326          0x80000001  2002::/64
 2.2.2.2         326          0x80000001  2001::/64

                 Link (Type-8) Link States (Area 1)

ADV Router       Age          Seq#        Link ID    Interface
 3.3.3.3         215          0x80000001  4          Gi0/2
 2.2.2.2         326          0x80000001  4          Gi0/1
 3.3.3.3         221          0x80000001  3          Gi0/1

                 Intra Area Prefix Link States (Area 1)

ADV Router       Age          Seq#        Link ID    Ref-lstype  Ref-
LSID
 2.2.2.2         221          0x80000003  0          0x2001      0
 2.2.2.2         221          0x80000001  4096       0x2002      4
 3.3.3.3         203          0x80000004  0          0x2001      0
```

(##This is like a Type 1 LSA for OSPFv2.)
(##This is like a Type 2 LSA for OSPFv2.)
(##This is like a Type 3 LSA for OSPFv2.)
(##Type 8 LSAs are going to provide information about the IPv6 addresses on a link, including any link-local addresses.)
(##These are Type 9 LSAs. They provide information for stub networks and transit networks and they are only sent within an area.)

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